MIL-E-1/1731A
18 April 1974
SUPERSEDING
MIL-E-1/1731
12 February 1973

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE DOD-020 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Passive TR-limiter, frequency range of 16,160 to 16,760 MHz, peak power 12.5 kw

ABSOLUTE RATINGS:

Parameter:	Incident power	tp	prr	Temperature Operating	Temperature Nonoperating	Alt
Unit:	kw	μs	pps	°C	°C	ft
Maximum:	12.5	2.0	4,000	+71	+100	70,000
Minimum:	--	0.15	200	-55	-65	---

PHYSICAL CHARACTERISTICS:

Dimension: See figure 1
Mounting position: Any
Weight: 2.5 ounces, maximum
Material and finishes: See figure 1

TEST CONDITIONS:

Parameter:	Incident power	tp	prr	F
Unit:	kw	μs	pps	MHz
Tolerance:	±10%	0.1	---	±3%
	10	1.0	1,000	F3

Test	Frequencies	
F	MHz	±
1	16,160	0.1%
2	16,310	0.1%
3	16,460	0.1%
4	16,610	0.1%
5	16,760	0.1%

GENERAL:

Qualification - Required

1/ Replaces the following part numbers

Texas Instruments Part No. 536609-1 and 523751-2
Microwave Associates Part No. MA-362E and MA-3887Z
Varian Associates Part No. BTR-608 and VDU-1029
Micro Dynamics, Inc. Part No. MD-70K8 and MD-80K15

DOD-020

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1					
4473	Low-level VSWR	2	F = F1, F2, F3, F4 and F5		---	1.3	---
4416	Insertion loss	2	F = F1, F2, F3, F4 and F5	Li	---	1.0	dB
4452	Flat-leakage power	-		pf	---	100	mw
4452	Spike-leakage energy	3		Ws	---	0.1	erg
4471	Recovery time	-		t	---	0.8	μs
4496	Firing power	-		Po/Du	---	200	mw
----	Phase shift	4	See figure 2	---	---	±2.5	Degrees
1027	Temperature cycling	10	1 cycle; TA = -55° to +71°C	---	---	---	---
	Quality conformance inspection, part 2						
4003	Pressurizing	10	20 psia	---	---	---	---
4101	Dielectric material strain	5, 10		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
----	Life test	7	Group D; TA = -55° to +71°C cycled continuously	t	1,000	---	hrs
----	Life-test end points:	-					
----	Crystal degradation	6, 7		ΔNF	---	1	dB
4471	Recovery time	-		t	---	1.0	μs
4452	Flat-leakage power	-		pf	---	100	mw
4452	Spike-leakage energy	3		Ws	---	0.1	erg
4416	Insertion loss	2	F = F1, F2, F3, F4 and F5	Li	---	1.0	dB
	Periodic-check tests						
1042	Shock, specified pulse	8	Test condition A	---	---	---	---
1032	Vibration, mechanical	9		---	---	---	---
1027	Temperature cycling life test	10	10 cycles (min)	---	---	---	---
1011	Humidity	10, 11		---	---	---	---

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection part 1, shall be 1.0 inspection level II.

2. A swept-frequency method may be used for this test.

3. The maximum spike amplitude shall not exceed 5 watts (peak) over the frequency range of F1 through F5.

4. Phase measurements referenced to a 0.807 inch (20.50 mm) length of 0.311 by 0.622 inch (7.90 by 15.80 mm) size guide as a blank.

5. Only the input window shall be immersed in water during this test.

6. The TUT shall be terminated with a type 1N5764, or equal diode in a balanced mixer. At the beginning of the life test, the crystal noise figure (NF) shall meet the minimum requirements for the specified crystals under the standard specified conditions. The initial crystal noise figure shall be measured and recorded. The crystal noise figure shall be remeasured periodically during the course of life testing. Life test shall be terminated and the TUT shall have failed life test when the crystal noise figure degradation (ΔNF) from the initial value has exceeded the specified limit.

7. One crystal failure will be allowed during the life test except that no failures for noise figure degradation shall be allowed. If a crystal failure occurs, a new crystal shall be inserted in the mount and the life test shall continue for a minimum of 200 hours, even if the life test is extended beyond the required 1,000 hours.

8. The TUT shall be shocked 5 times in the direction shown in figure 1 when actual mounting conditions are simulated. The TUT shall be capable of satisfying all requirements of this specification without damage or failure before and after being subjected to the specified shock test.

9. Vibration. The TUT shall be capable of satisfying all requirements of this specification without damage or failure before and after being subjected to the specified vibration test with the amplitude as specified herein.

Frequency (HZ)		Amplitude (Inches)
5 to 15	0.075 ±.012	(0.150 Total displacement)
16 to 25	0.050 ±.008	(0.100 Total displacement)
26 to 33	0.025 ±.004	(0.050 Total displacement)

10. After completion of this test, the tube shall meet the requirements of Method 4496, Firing power.

11. The output port shall be sealed with a gasket and cover plate to prevent moisture from entering the limiter section during this test.

Custodians:
Army - EL
Navy - EC
Air Force - 85

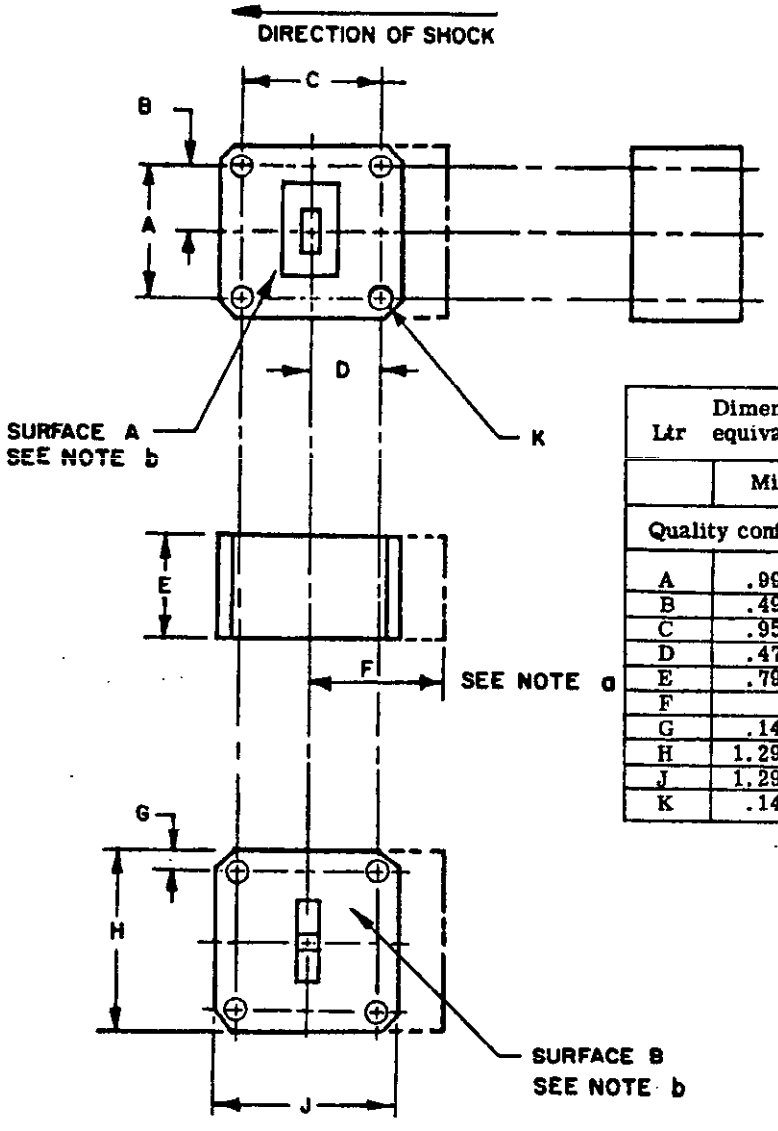
Review activities:
Army - MU, ME
Air Force - 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11, 84

Preparing activity:
Air Force - 85

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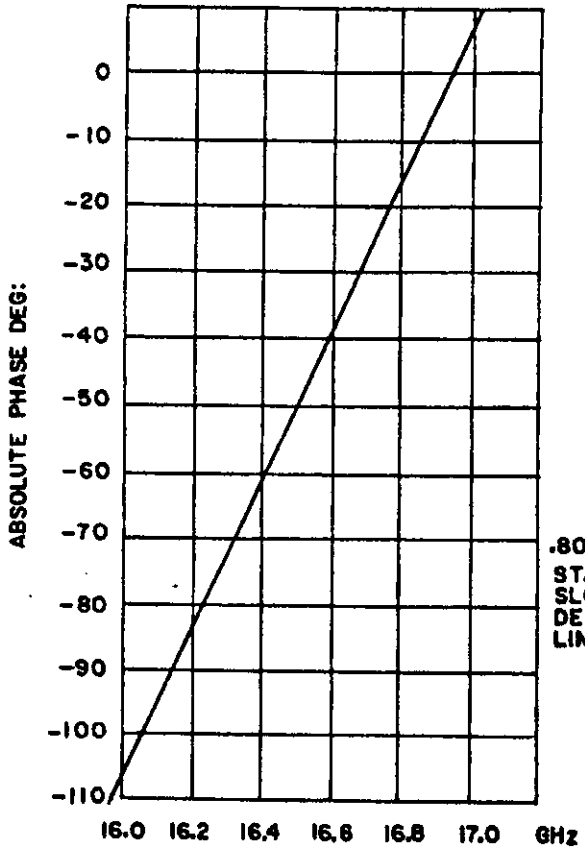


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	.990(25.15)	.998(25.35)
B	.493(12.52)	.501(12.73)
C	.952(24.19)	.960(24.38)
D	.474(12.04)	.482(12.24)
E	.797(20.24)	.817(20.75)
F		1.10 (27.94)
G	.140 (3.56)	.160 (4.06)
H	1.296(32.92)	1.328(33.73)
J	1.296(32.92)	1.328(33.73)
K	.147(3.73) Dia clearance 4 places	

NOTES:
a. Any projections from the main body of the device shall be in the areas described by the phantom lines, or .25 inch maximum from any other side excluding faces.
b. Surfaces A and B shall be compatible with aluminum flanges per dissimilar metals requirement of MIL-STD-889. All other surfaces shall be painted a dull black.

FIGURE 1. Outline drawing of electron tube type DOD-020

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FREQUENCY (GHz)	PHASE (Degrees)
16.16	88.5
16.20	84.0
16.50	50.0
16.70	27.5
16.76	20.5

FIGURE 2. Phase shift.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-1/1731A Electron Tube, Gas Switching Type DOD-020 1/

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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DD FORM 1 JAN 72 1426

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